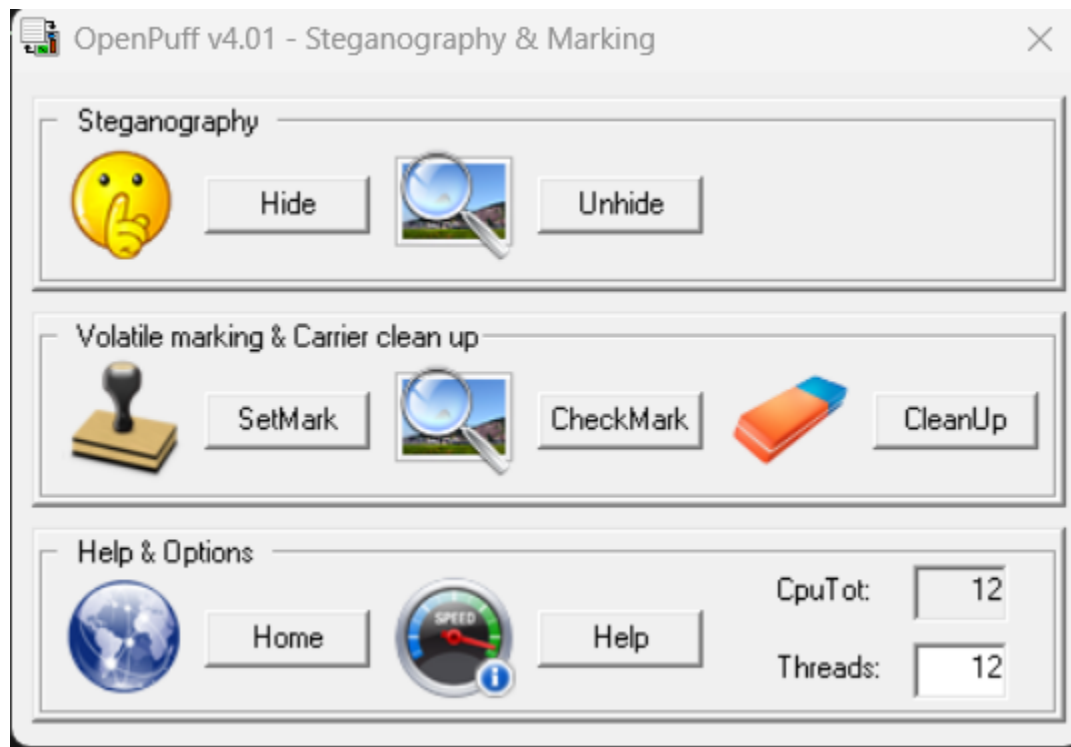


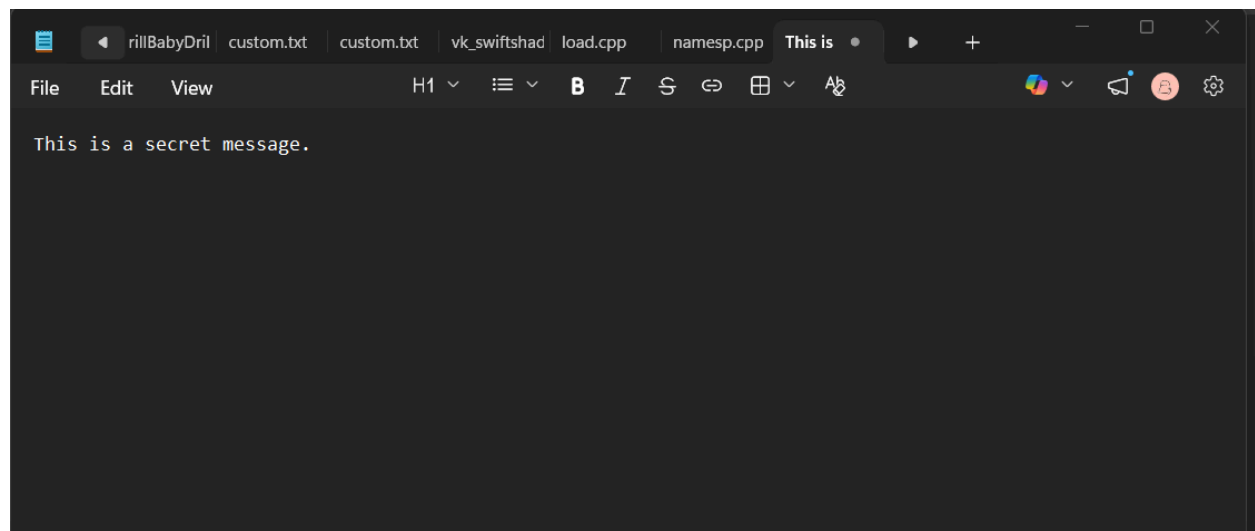
Đoàn Việt Đức - 23020599

Project 6-1: Using OpenPuff Steganography

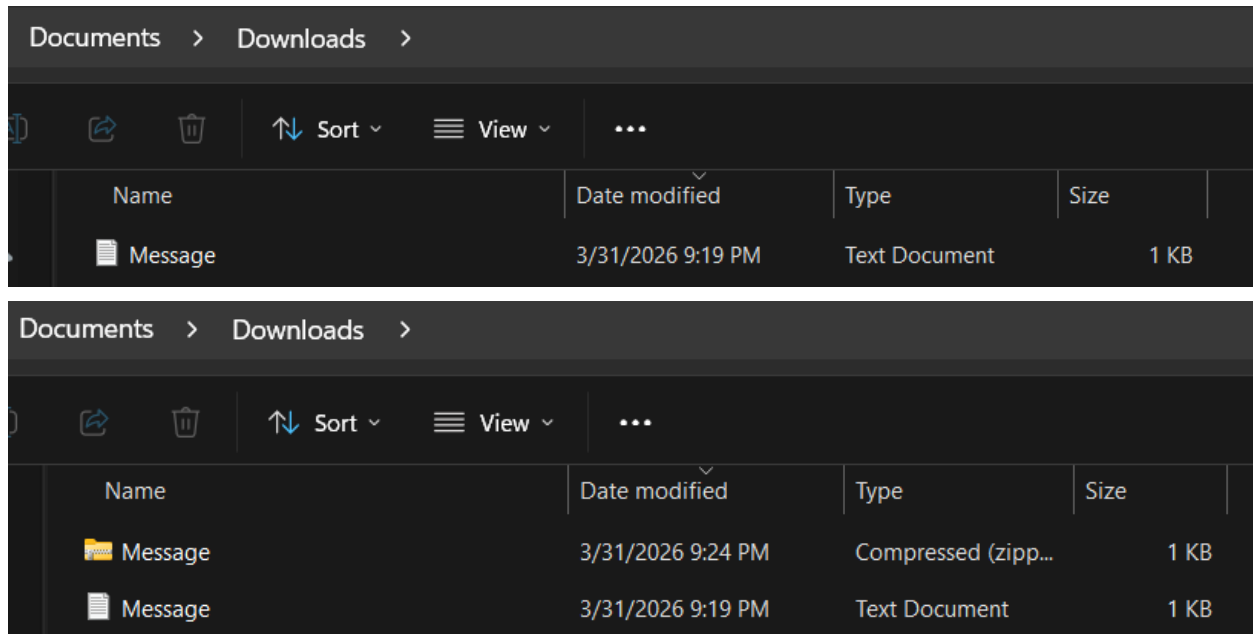
-OpenPuff Installation and Extraction:



-Create a secret message with notepad:

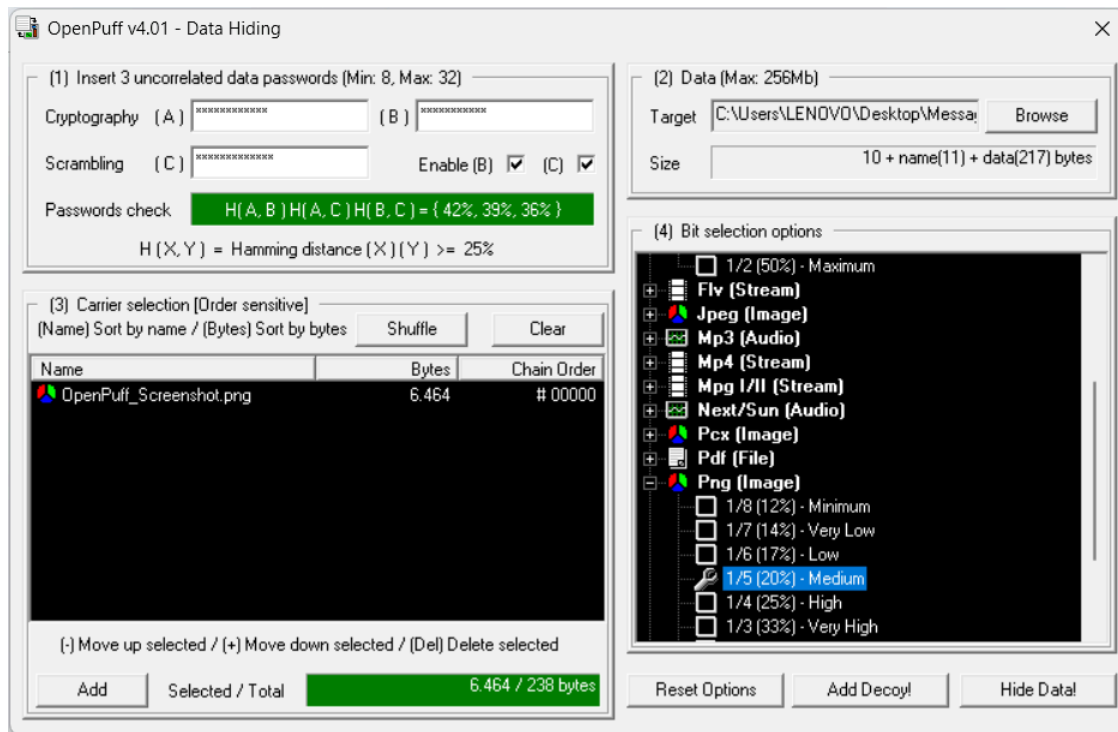


-Save as Message.txt and zip it:

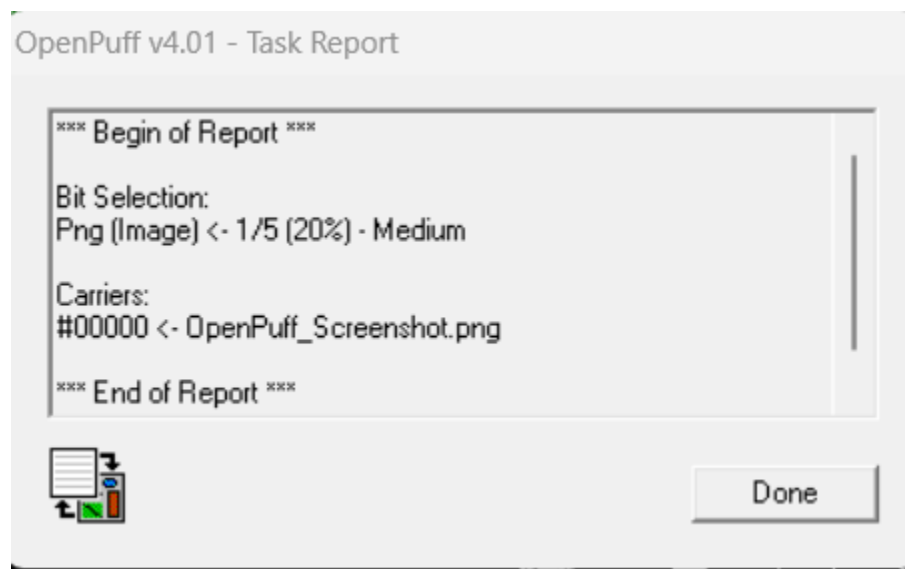
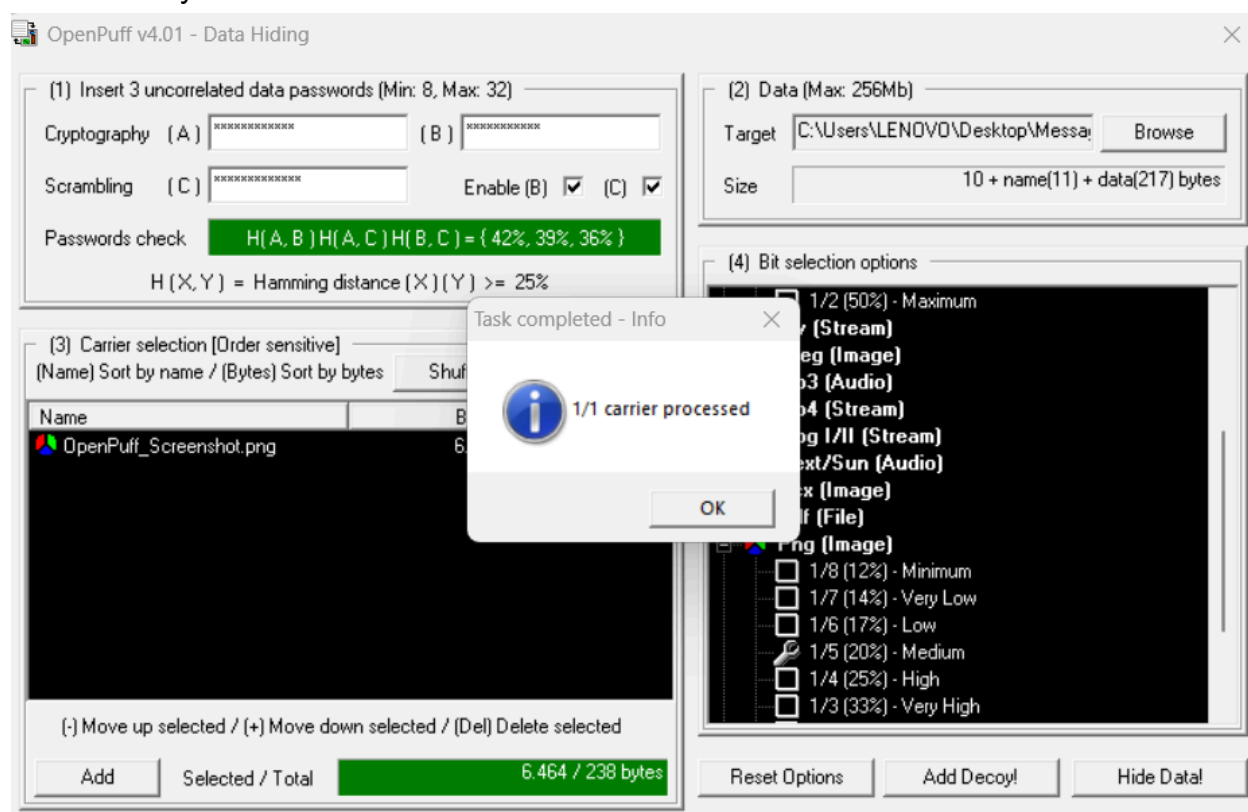


-Create three random passwords for **Cryptography (A)**, **Cryptography (B)** and **Scramble (C.)** Carrier: OpenPuff_Screenshot.jpg. Data: Message.zip

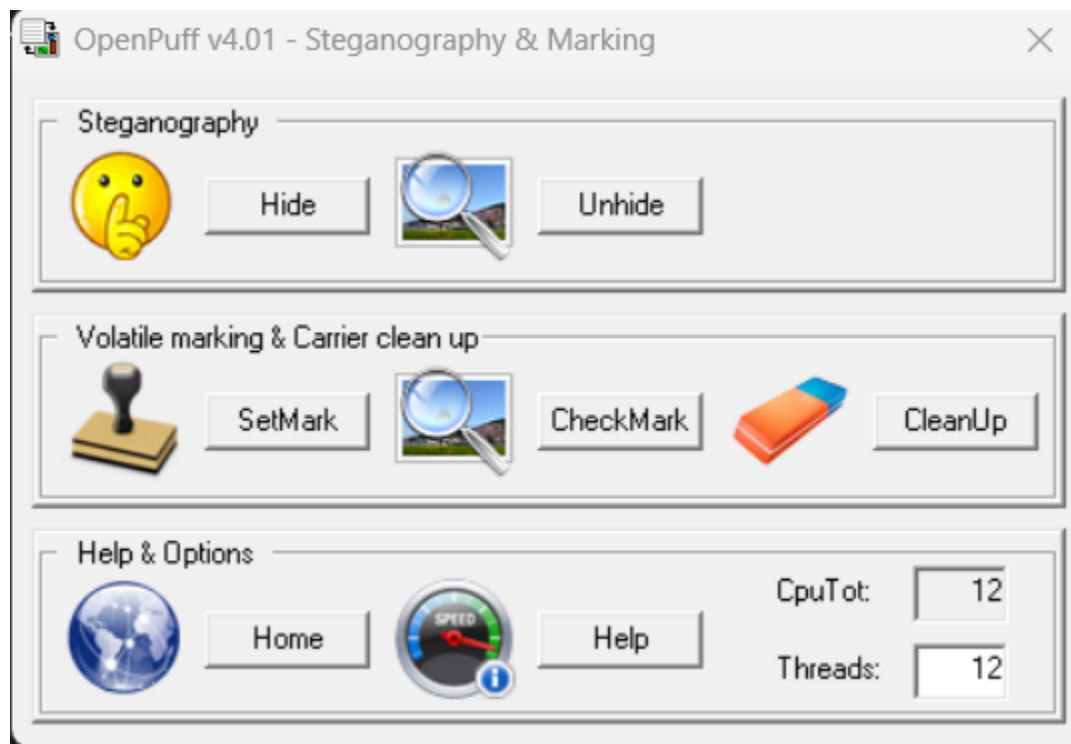
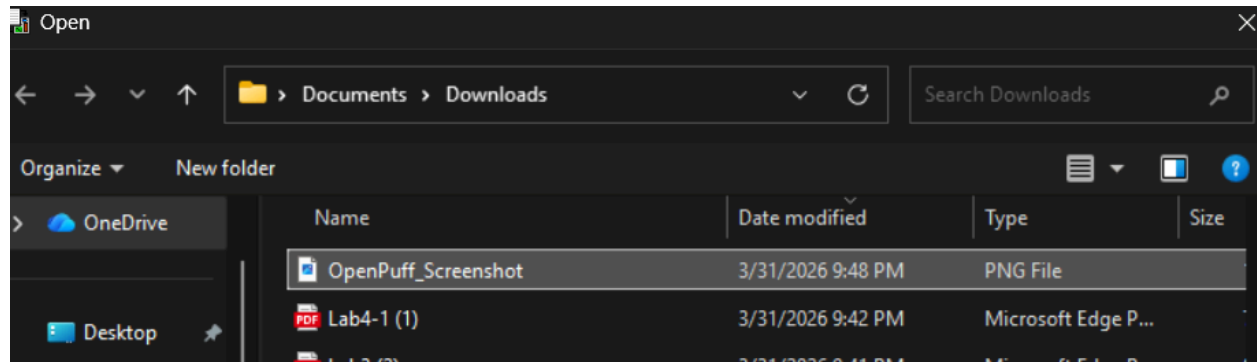
- Casternia65*
- DvD0505019*
- Potato04201~*



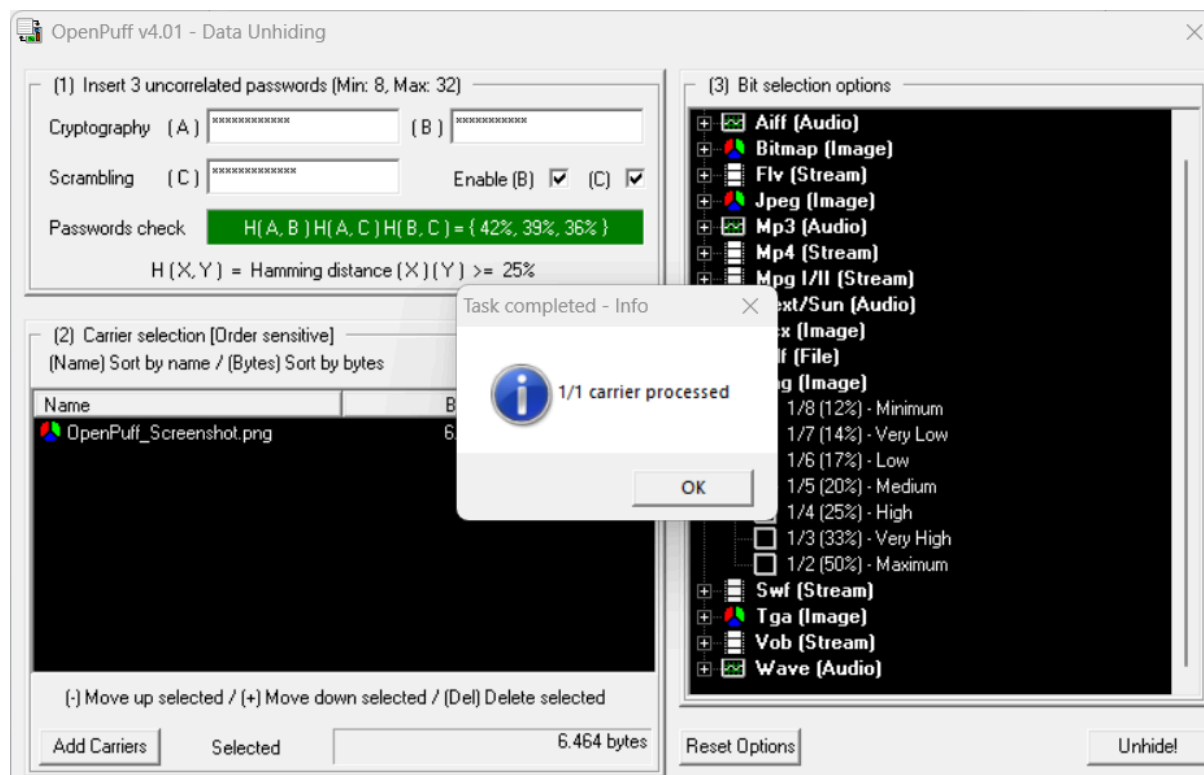
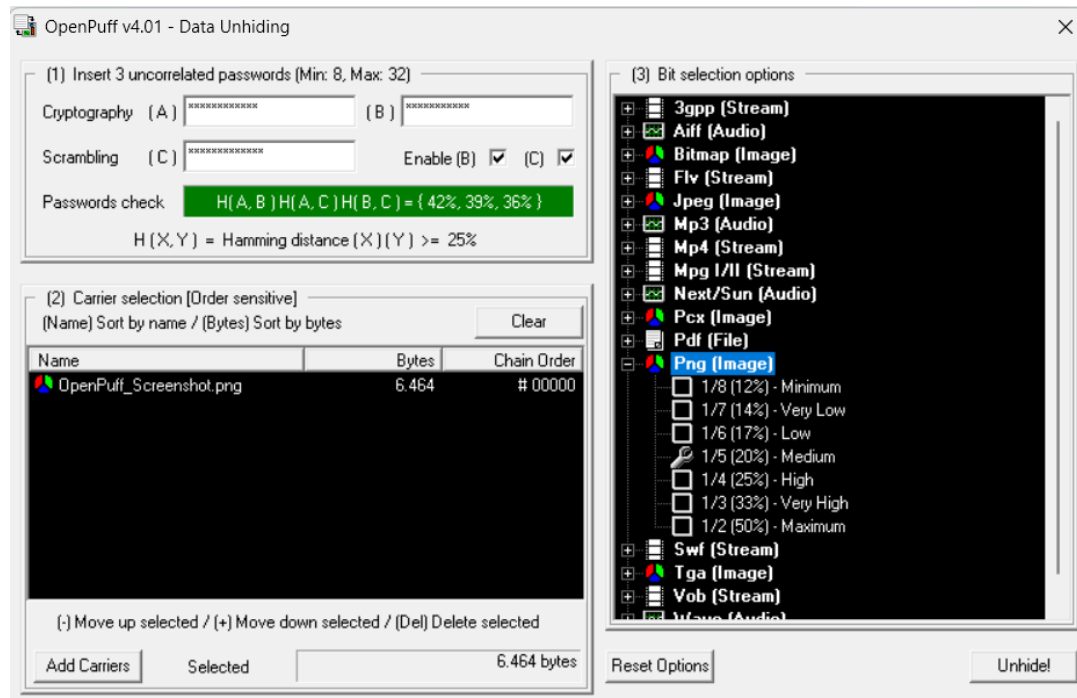
-Successfully hide data:

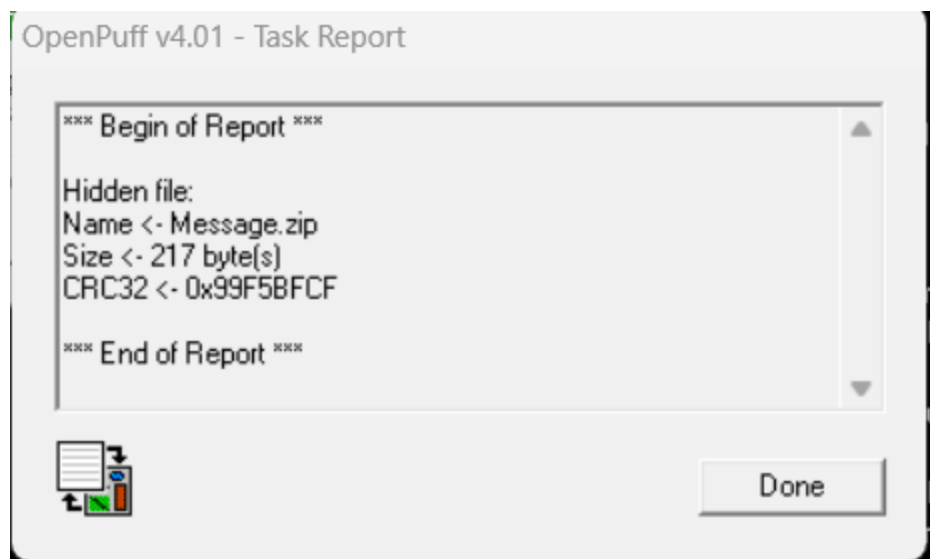


-The carrier file (OpenPuff_Screenshot.jpg) after looks completely normal:

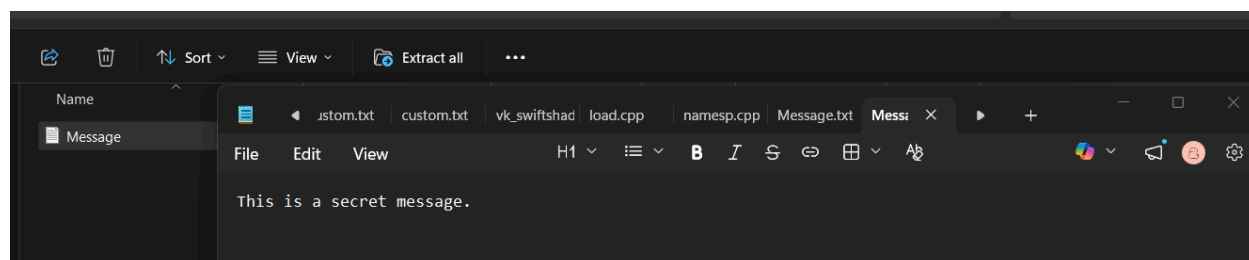
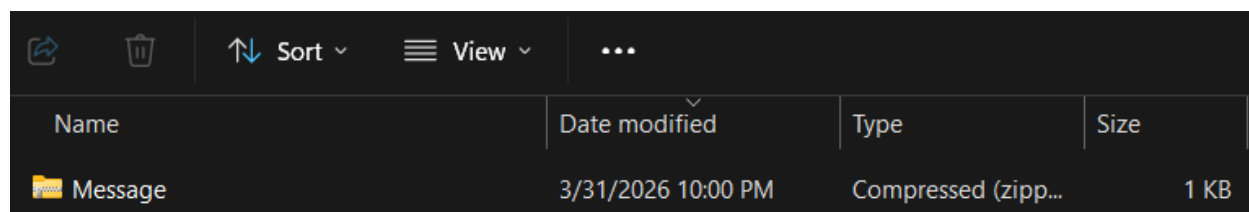


-Uncovering the message from the carrier file:

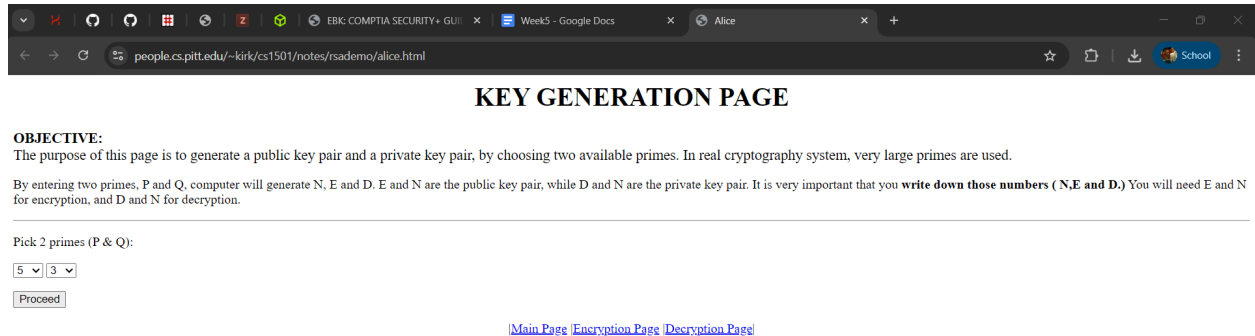




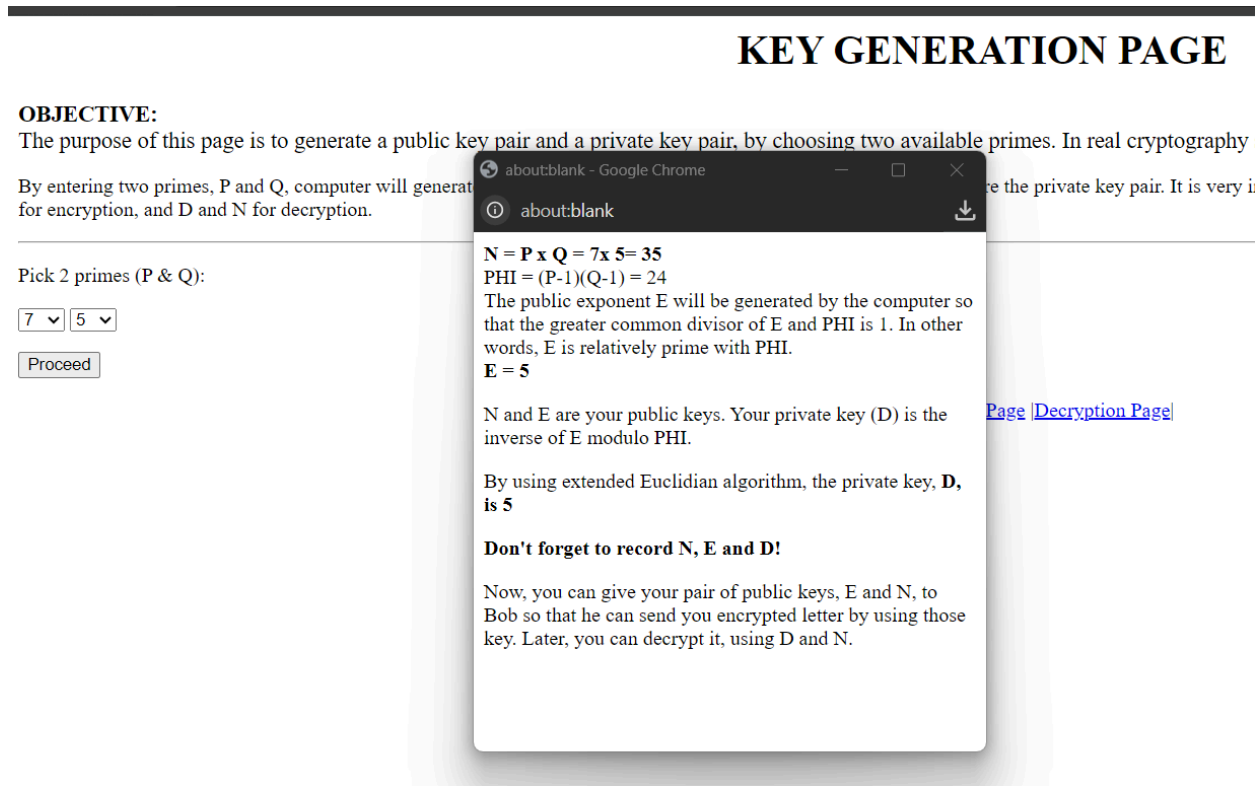
-Getting the unhided zip file: **(Message is successfully decoded)**



Project 6-2: Running an RSA Cipher Demonstration



The screenshot shows a web browser window with the address bar displaying `people.cs.pitt.edu/~kirk/cs1501/notes/rsademo/alice.html`. The page title is "KEY GENERATION PAGE". The "OBJECTIVE:" section explains the purpose of the page: to generate a public key pair and a private key pair by choosing two available primes. It states that in real cryptography, very large primes are used. The instructions mention that by entering two primes, P and Q, the computer will generate N, E and D. E and N are the public key pair, while D and N are the private key pair. It emphasizes that it is very important to write down the numbers (N, E and D) as they will be needed for encryption and decryption. Below the objective, there is a section "Pick 2 primes (P & Q):" with two dropdown menus. The first dropdown is set to "5" and the second is set to "3". A "Proceed" button is located below the dropdowns. At the bottom of the page, there are three links: [Main Page](#), [Encryption Page](#), and [Decryption Page](#).



The screenshot shows the same "KEY GENERATION PAGE" as above, but with a modal window open in the center. The modal window has a title bar that says "about:blank - Google Chrome" and a URL bar that says "about:blank". The content of the modal window is as follows:

N = P x Q = 7 x 5 = 35
PHI = (P-1)(Q-1) = 24
The public exponent E will be generated by the computer so that the greater common divisor of E and PHI is 1. In other words, E is relatively prime with PHI.
E = 5

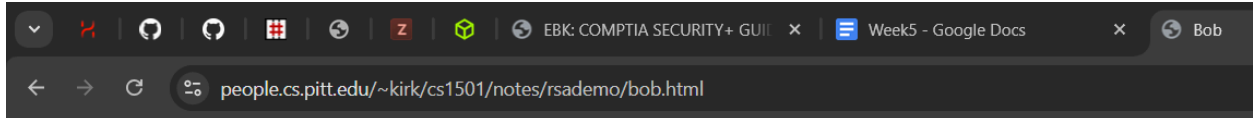
N and E are your public keys. Your private key (D) is the inverse of E modulo PHI.

By using extended Euclidian algorithm, the private key, **D**, is **5**

Don't forget to record N, E and D!

Now, you can give your pair of public keys, E and N, to Bob so that he can send you encrypted letter by using those key. Later, you can decrypt it, using D and N.

The background page is partially obscured by the modal window. The "Pick 2 primes (P & Q):" section is visible, showing the dropdown menus set to "7" and "5", and the "Proceed" button. The links [Main Page](#), [Encryption Page](#), and [Decryption Page](#) are also visible at the bottom of the page.



ENCRYPTION PAGE

OBJECTIVE:

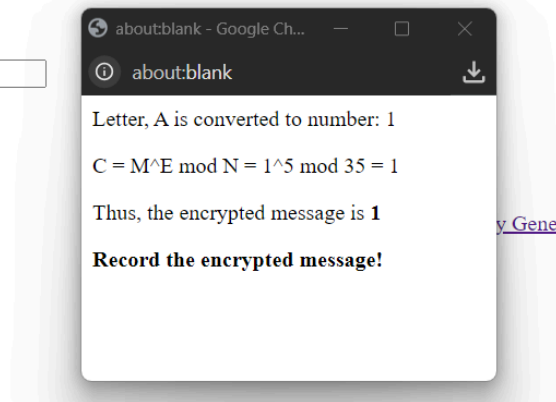
The purpose is to encrypt a letter using RSA algorithm. Knowing E and N is required.

Record the encrypted message!

Pick a letter to cipher:

Enter Alice' Exponent key, E:

Enter Alice' N value:



[Key Generation Page](#) [Decryption Pa](#)

DECRYPTION PAGE

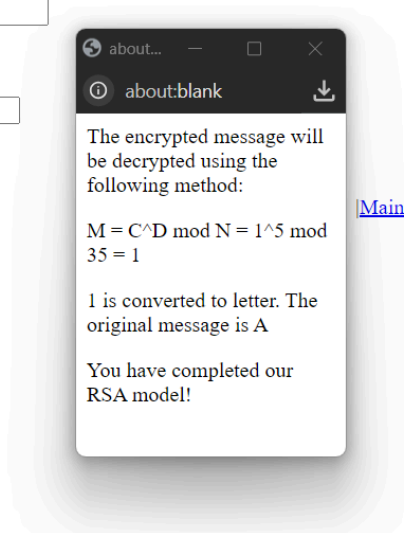
OBJECTIVE:

This is the final step of understanding RSA algorithm. The purpose is to decrypt the encrypted message, by knowing the private key.

Enter the encrypted message:

Enter your N value:

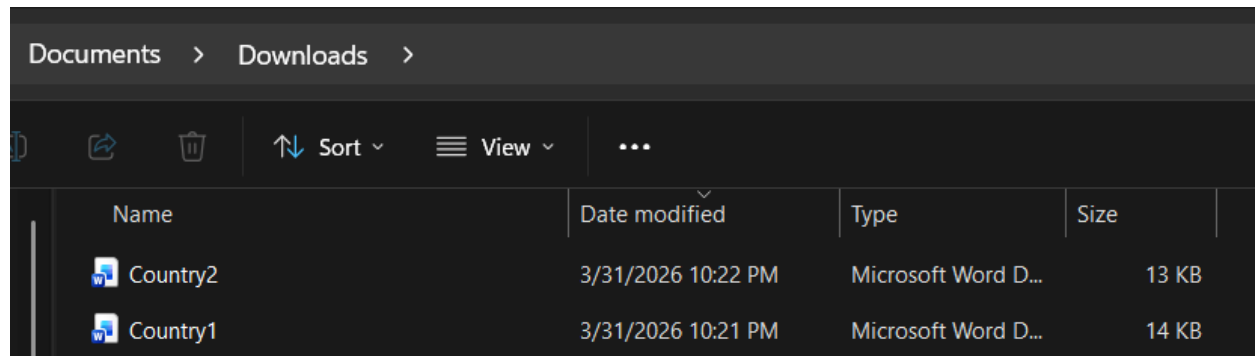
Enter your private key, D:



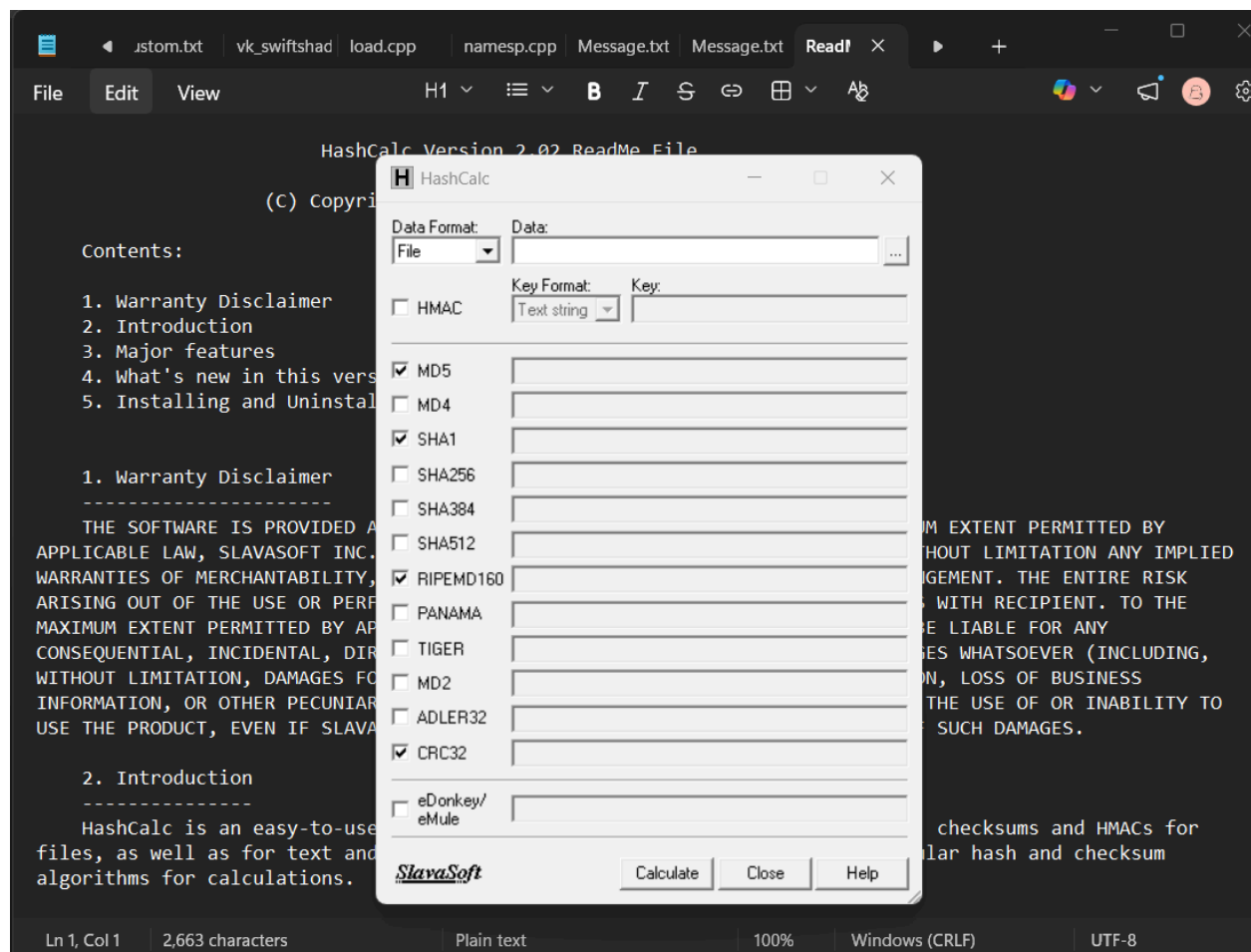
[Main Page](#) [Key Generation Page](#) [Encryption Page](#)

Project 6-3: Installing GUI Hash Generator and Comparing Digests

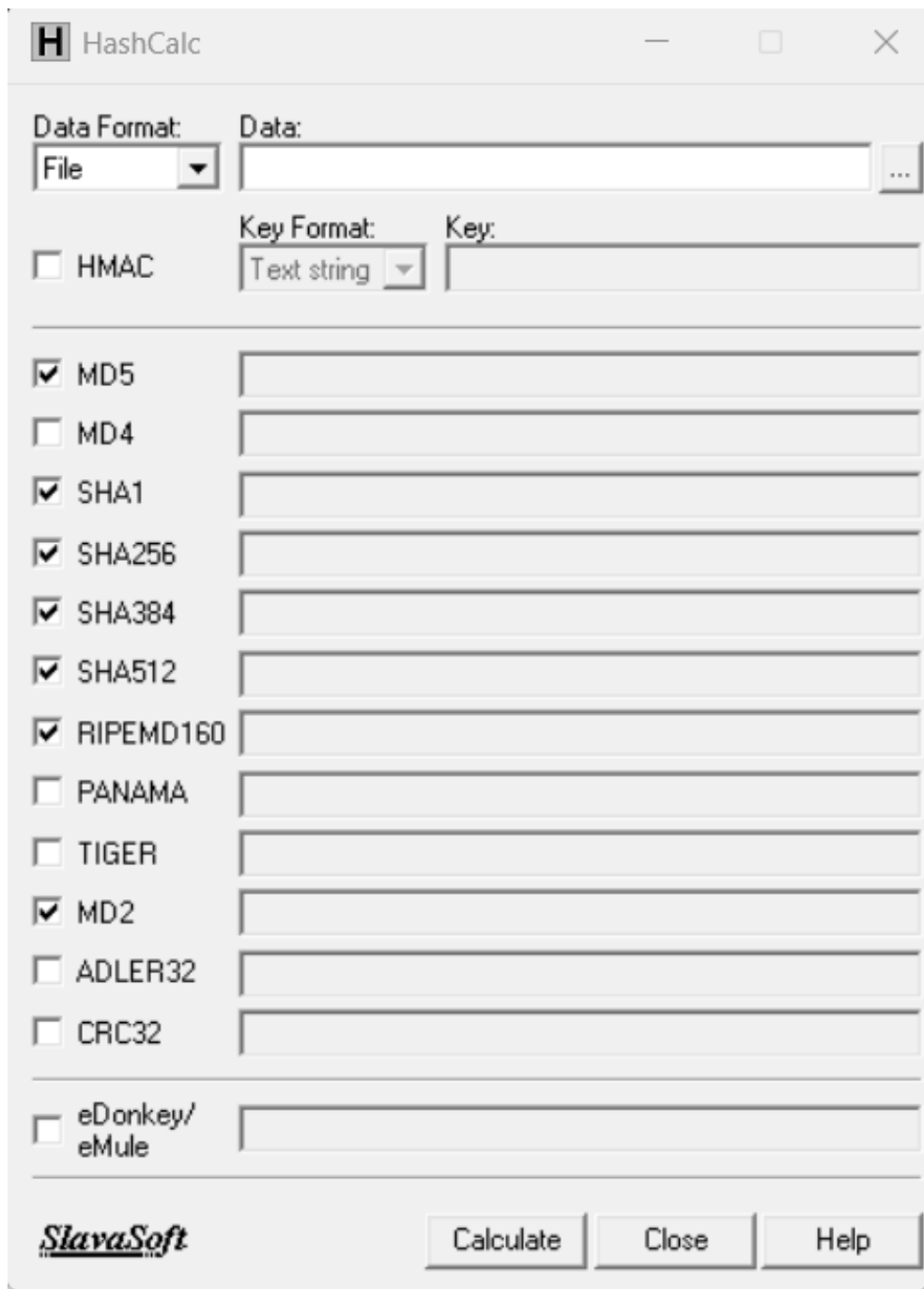
-Create **Country1.docx** and **Country2.docx** as instructed:



-Downloaded HashCalc:



-In addition to the hash algorithms selected by default, check the box next to the following hash algorithms to add them: **MD5**, **SHA256**, **SHA384**, **SHA512**, and **MD2**:



The screenshot shows the HashCalc application window. At the top, there is a title bar with the 'H' icon and the text 'HashCalc'. Below the title bar, there are two main sections. The first section contains a 'Data Format' dropdown menu set to 'File' and a 'Data' text input field with a browse button ('...'). The second section contains a 'Key Format' dropdown menu set to 'Text string' and a 'Key' text input field. Below these sections, there is a list of hash algorithms, each with a checkbox and a corresponding output field. The algorithms and their selection status are: MD5 (checked), MD4 (unchecked), SHA1 (checked), SHA256 (checked), SHA384 (checked), SHA512 (checked), RIPEMD160 (checked), PANAMA (unchecked), TIGER (unchecked), MD2 (checked), ADLER32 (unchecked), CRC32 (unchecked), and eDonkey/eMule (unchecked). At the bottom of the window, there is a 'SlavaSoft' logo and three buttons: 'Calculate', 'Close', and 'Help'.

Algorithm	Selected
MD5	Yes
MD4	No
SHA1	Yes
SHA256	Yes
SHA384	Yes
SHA512	Yes
RIPEMD160	Yes
PANAMA	No
TIGER	No
MD2	Yes
ADLER32	No
CRC32	No
eDonkey/eMule	No

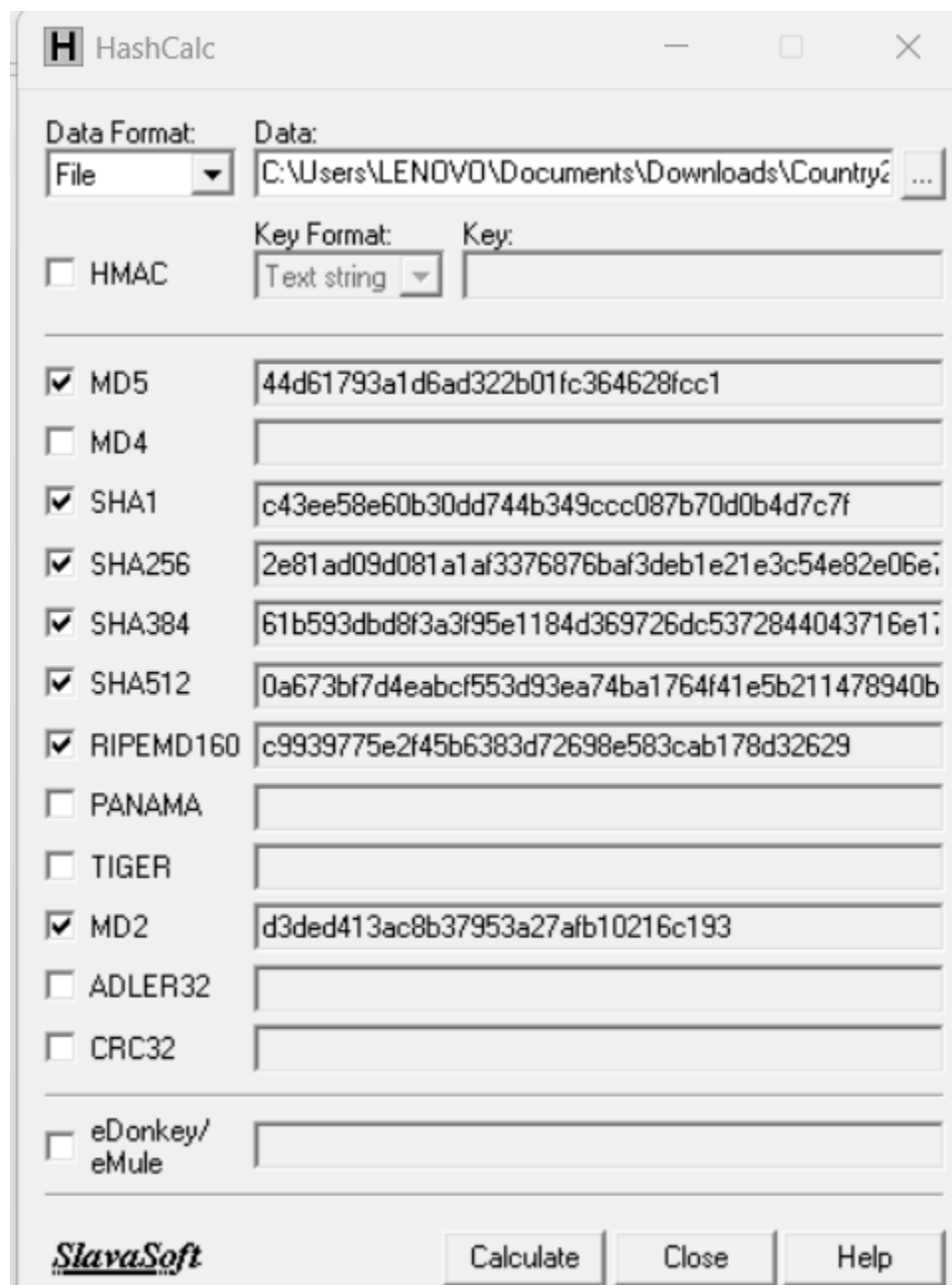
-Calculate the **Country1.docx** with HashCalc:

The screenshot shows the HashCalc application window. The 'Data Format' is set to 'File' and the 'Data' field contains the path 'C:\Users\LENOVO\Documents\Downloads\Country1'. The 'Key Format' is set to 'Text string' and the 'Key' field is empty. The 'HMAC' checkbox is unchecked. The following hash algorithms are checked and their corresponding hash values are displayed in the adjacent text boxes:

Algorithm	Hash Value
☒ MD5	4db4dcac8eeb44548f04daa216e34193
☐ MD4	
☒ SHA1	9bc2d6db17779b874fc212d39196c44d40194993
☒ SHA256	027144da825cb5dc537f7d4b0d35ef48a5767947d26b62t
☒ SHA384	8c6a0d567295cb646d547ca795a29d7293e98eedec263
☒ SHA512	bbf8f5a47793f985b8a23509a82addf4d32c60d6352d7f1d
☒ RIPEMD160	bf7bfcc47a6f62aca7a8814ef9c473d03a265609
☐ PANAMA	
☐ TIGER	
☒ MD2	06d3a59d1c91903f1e61055424e02bb3
☐ ADLER32	
☐ CRC32	
☐ eDonkey/ eMule	

At the bottom of the window, there is a 'SlavaSoft' logo and three buttons: 'Calculate', 'Close', and 'Help'.

-Calculate the **Country2.docx** with HashCalc:



-The digests are entirely different. Hashing algorithms take into account the entire data of a file it calculates. So changing any content, metadata, or information will alter the hash result for all the hashing algorithms, maintaining security integrity. **This also gives proof whether the file has been altered or not by comparing hashes.**

Project 6-4: Using Microsoft's Encrypting File System (EFS)

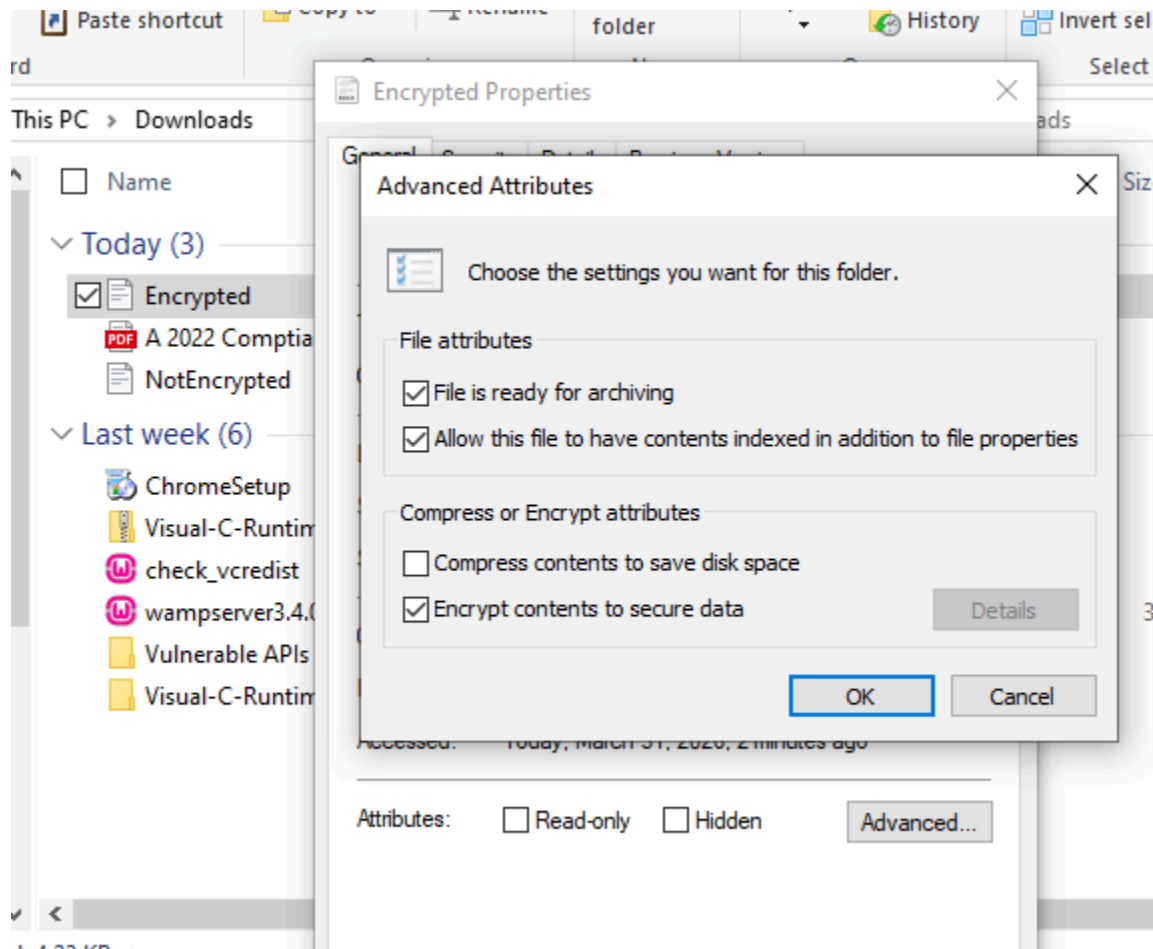
-Create files as required:

This PC > Downloads >

Search Downloads

☐ Name	Date modified	Type	Size
▼ Today (3)			
Encrypted	3/31/2026 9:00 AM	Office Open XML ...	5 KB
A 2022 Comptia Security+ Guide to N...	3/31/2026 8:59 AM	Microsoft Edge P...	13,389 KB
NotEncrypted	3/31/2026 9:00 AM	Office Open XML ...	5 KB

-Check **Encrypt contents to secure data**:



This document is now protected with EFS. All actions regarding encrypting and decrypting the file are transparent to the user and should not noticeably affect any computer operations.

Name	Date modified	type	Size
Today (3)			
☒ Encrypted	3/31/2026 9:00 AM	Office Open XML ...	5 KB
 A 2022 Comptia Security+ Guide to N...	3/31/2026 8:59 AM	Microsoft Edge P...	13,389 KB
 NotEncrypted	3/31/2026 9:00 AM	Office Open XML ...	5 KB

-The encrypted **Encrypted.docx** file takes longer to open than the **NotEncrypted.docx** file. Despite that, **the delay is not noticeable**.